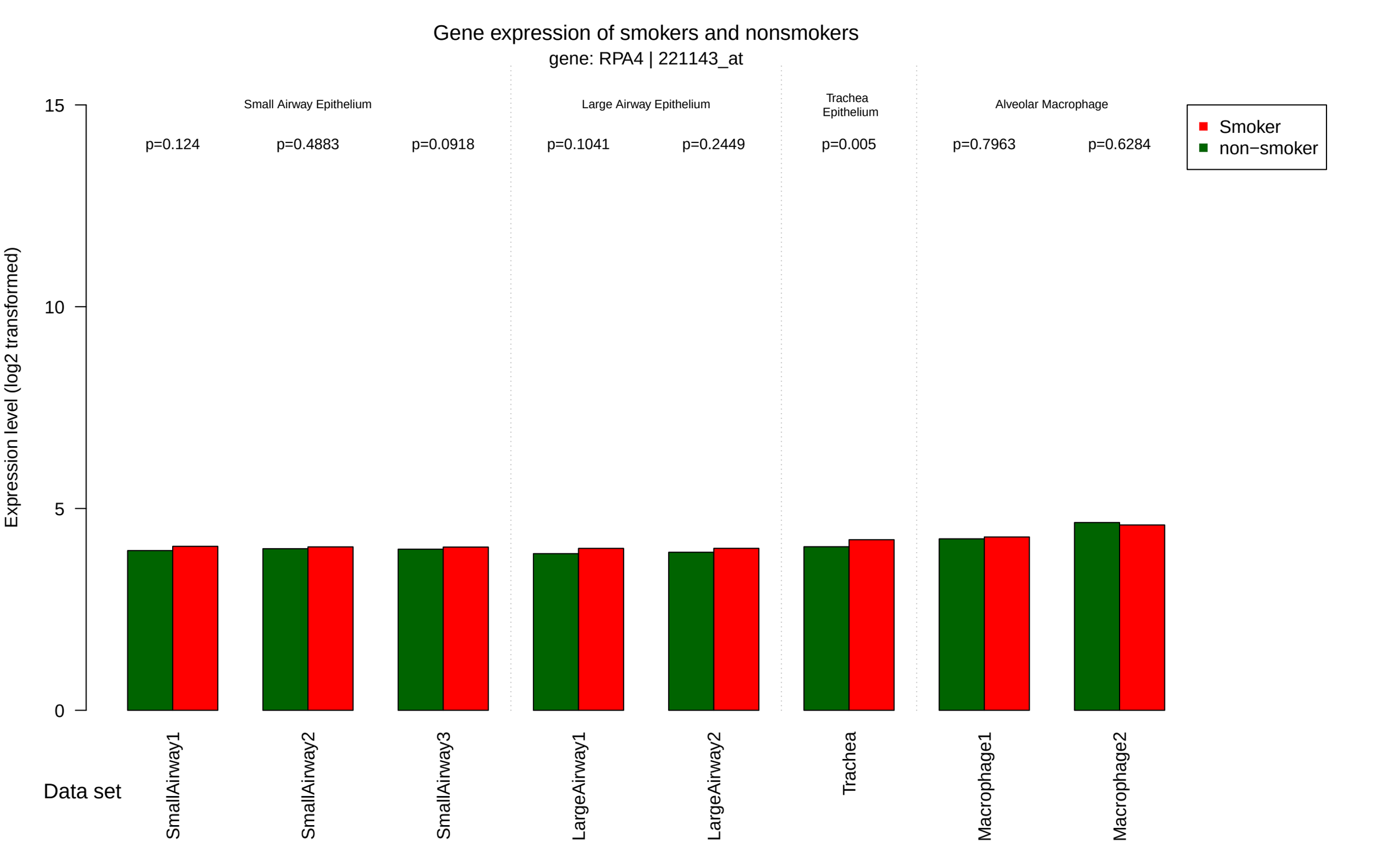
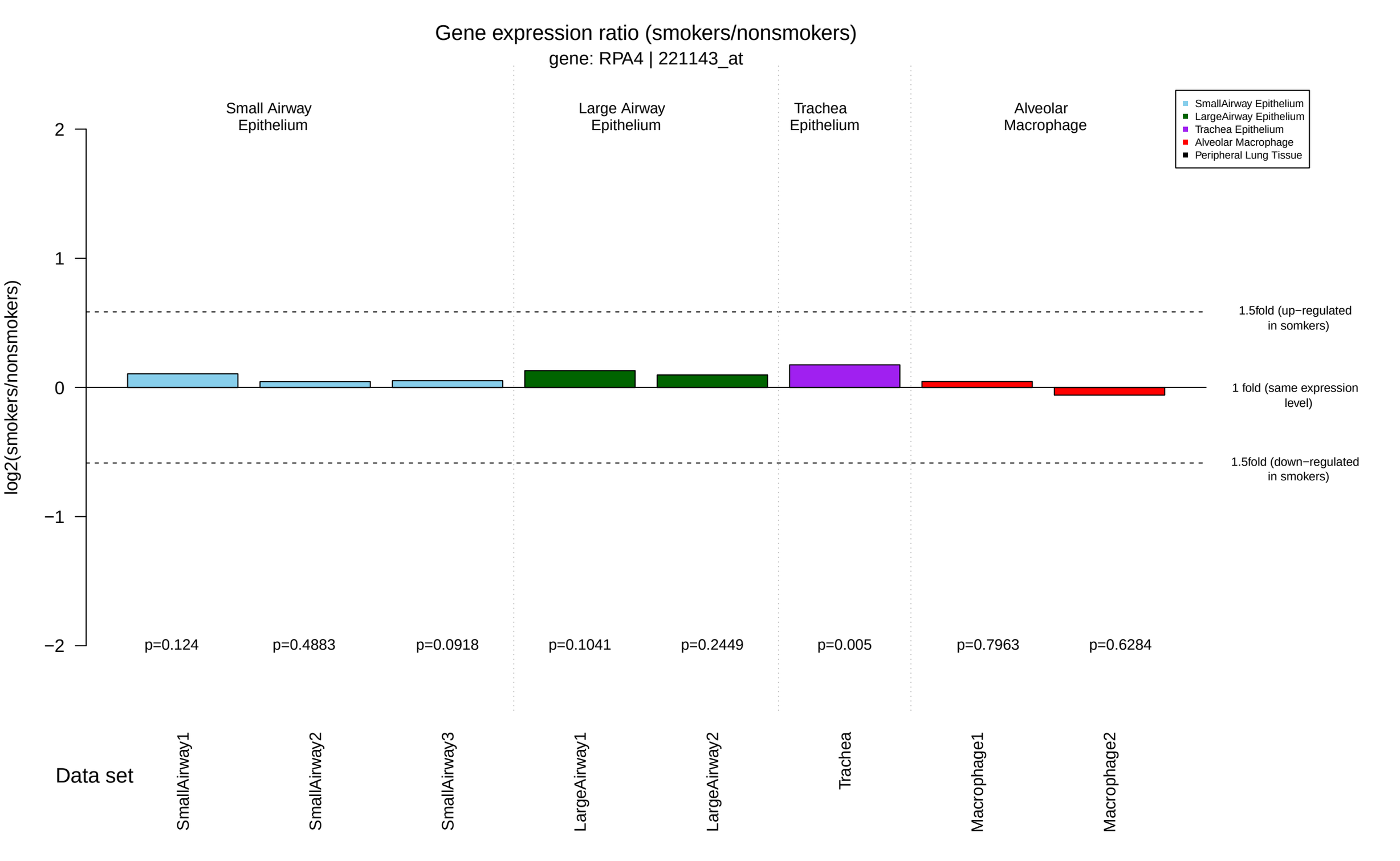
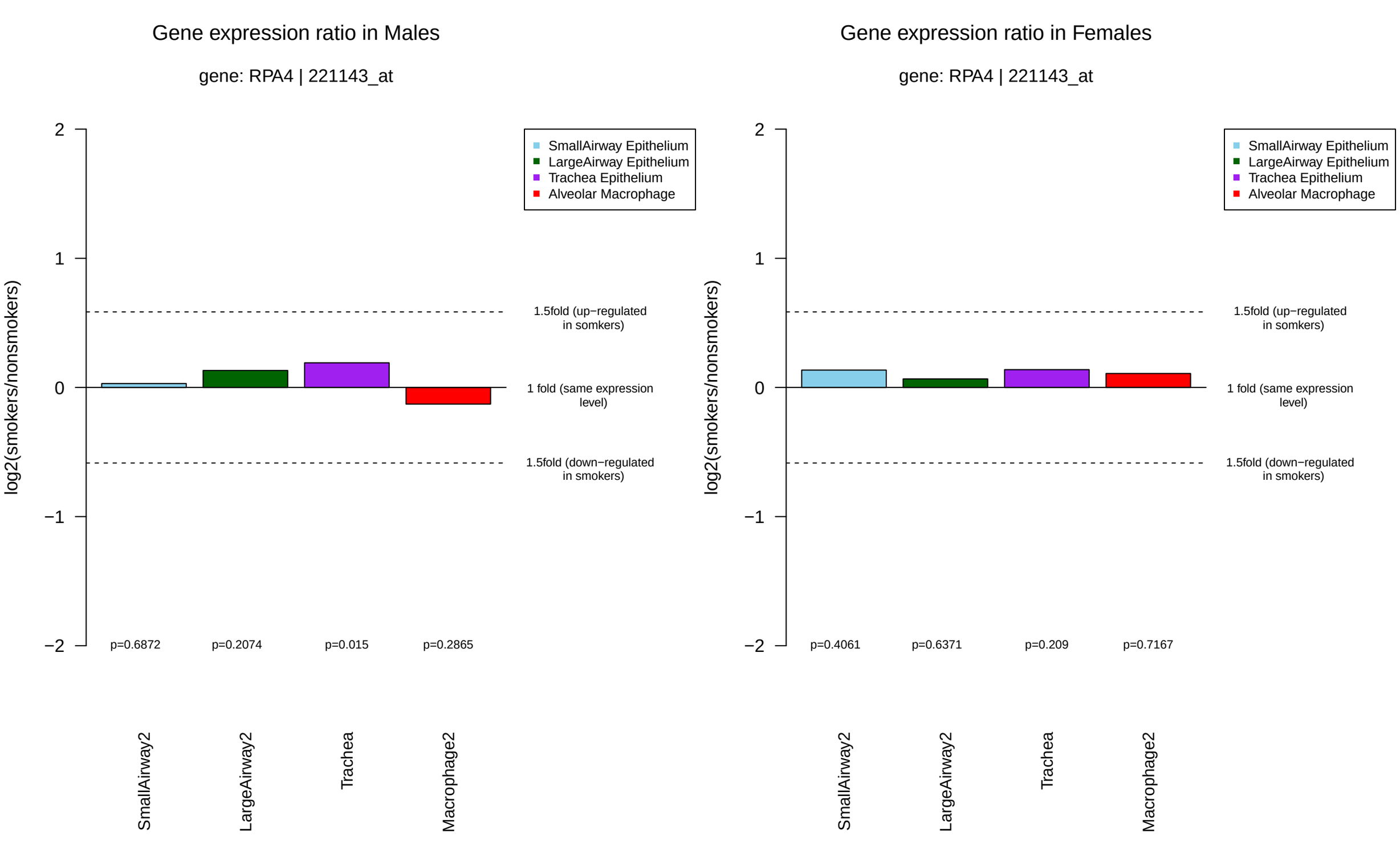
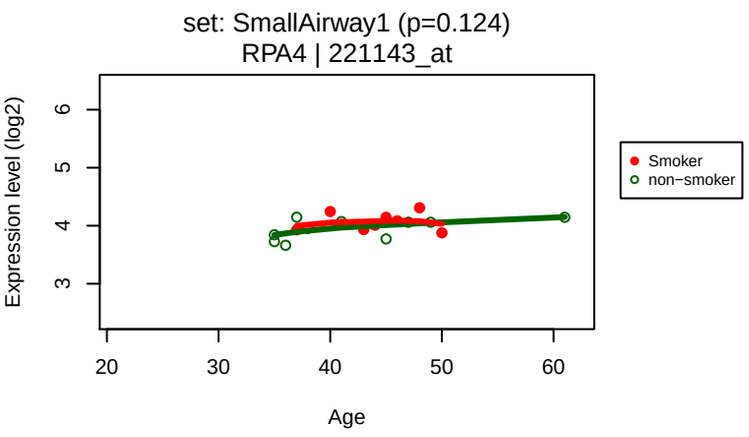
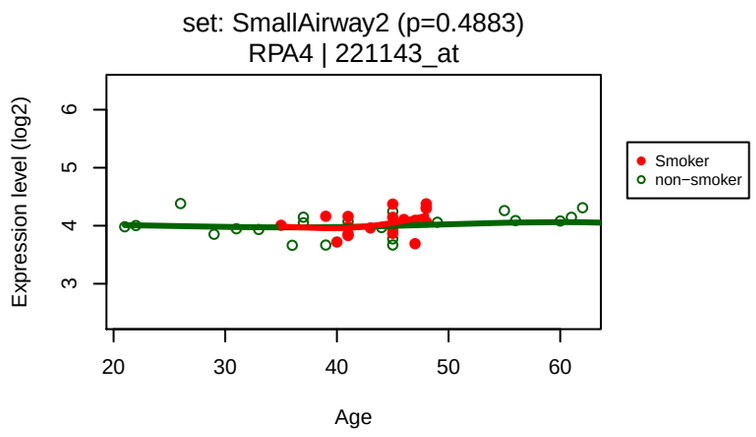
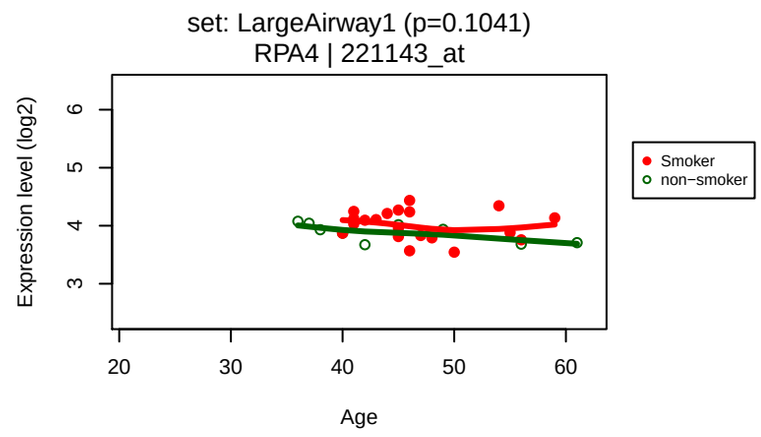

Gene expression of smokers and nonsmokers
gene: RPA4 | 221143_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.124
p=0.4883
p=0.0918
p=0.1041
p=0.2449
p=0.005
p=0.7963
p=0.6284
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

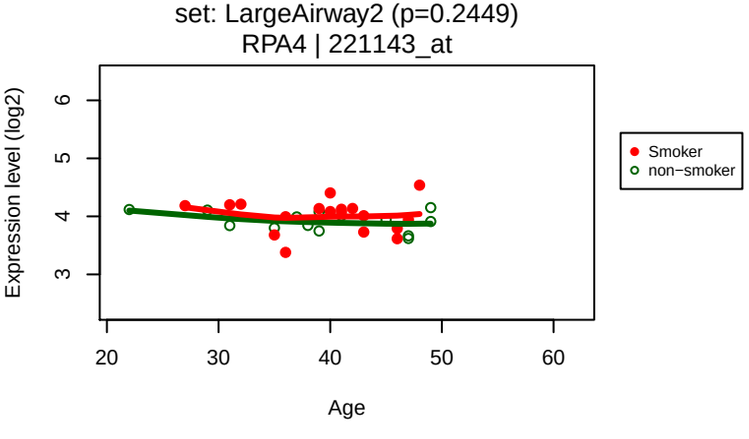

Gene expression ratio (smokers/nonsmokers)
gene: RPA4 | 221143_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.124
p=0.4883
p=0.0918
p=0.1041
p=0.2449
p=0.005
p=0.7963
p=0.6284
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

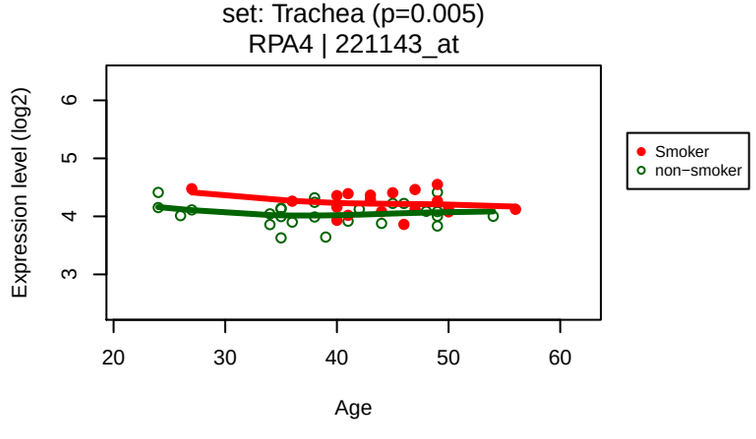

Gene expression ratio in Males
gene: RPA4 | 221143_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6872
p=0.2074
p=0.015
p=0.2865
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPA4 | 221143_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4061
p=0.6371
p=0.209
p=0.7167
SmallAirway2
LargeAirway2
Trachea
Macrophage2

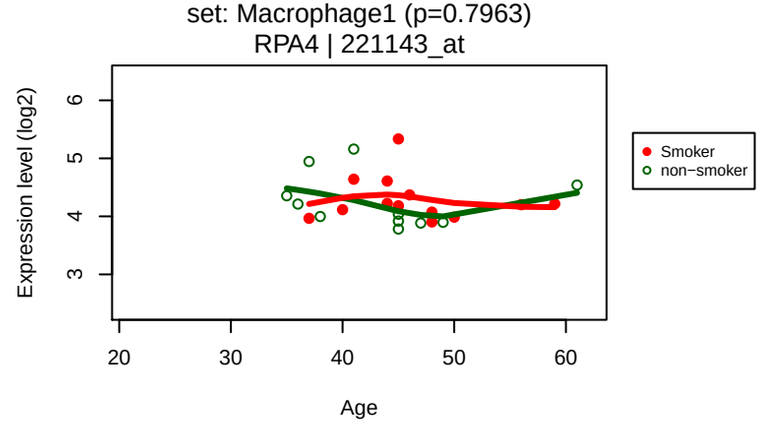

set: SmallAirway1 (p=0.124)
RPA4 | 221143_at
Expression level (log2)
Age
Smoker
non−smoker

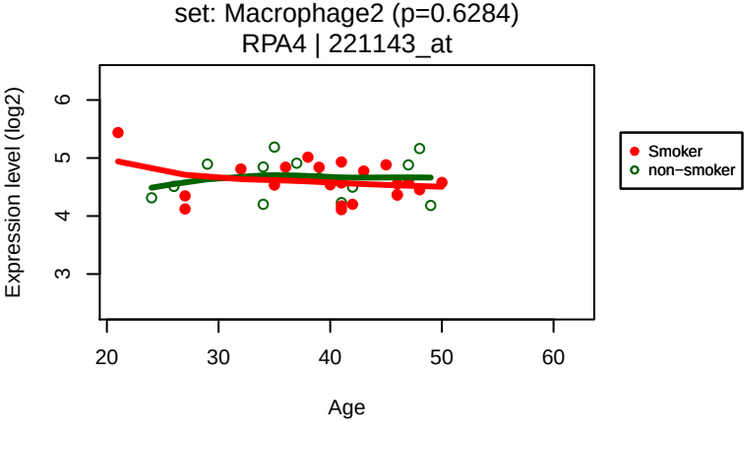

set: SmallAirway2 (p=0.4883)
RPA4 | 221143_at
Expression level (log2)
Age
Smoker
non−smoker

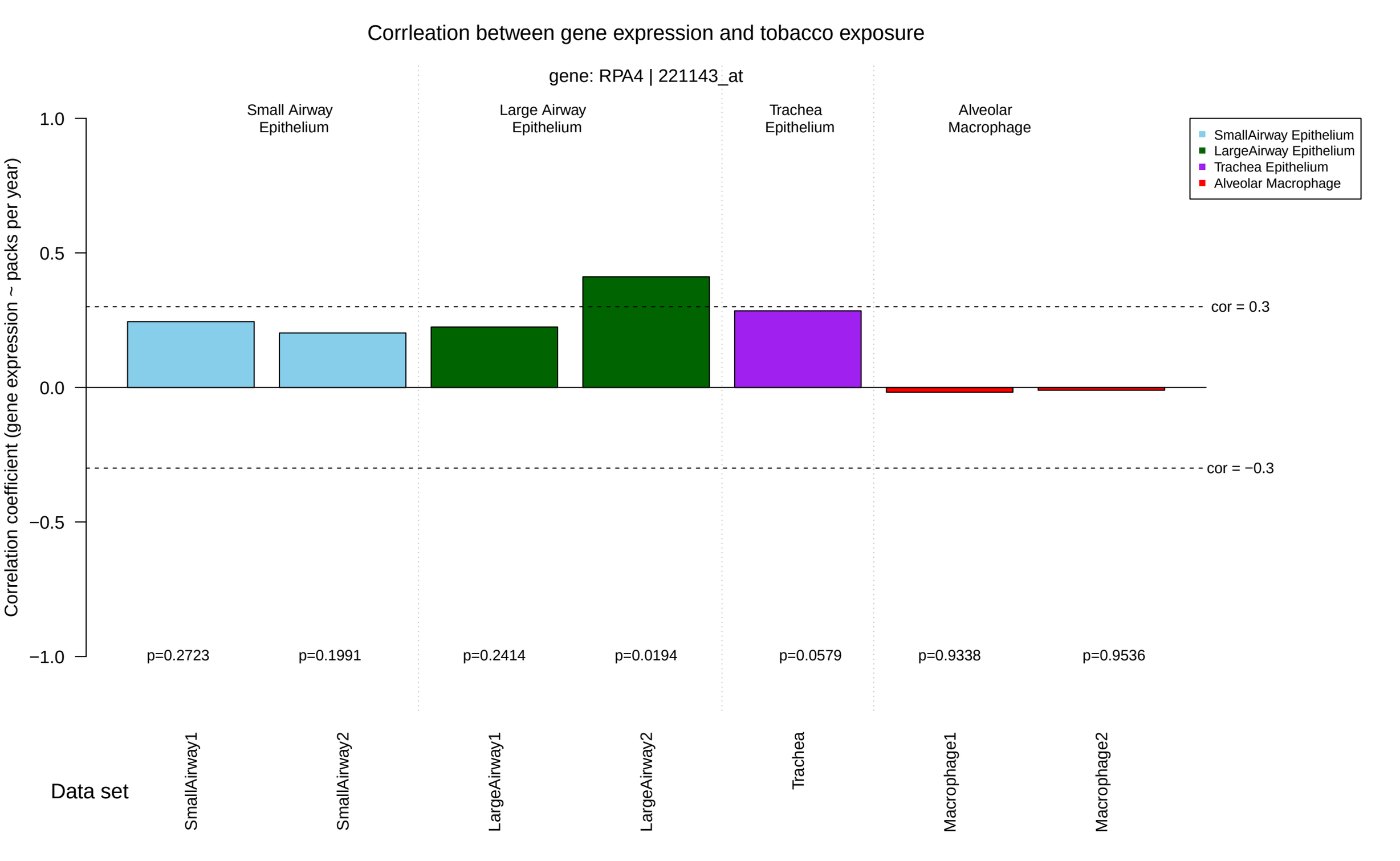

set: LargeAirway1 (p=0.1041)
RPA4 | 221143_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2449)
RPA4 | 221143_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.005)
RPA4 | 221143_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7963)
RPA4 | 221143_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6284)
RPA4 | 221143_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPA4 | 221143_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2723
p=0.1991
p=0.2414
p=0.0194
p=0.0579
p=0.9338
p=0.9536
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2